NICHOLAS CARRERA

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Recent DePaul graduate (B.S. UX Design, CS minor) who designs and builds interactive web, real-time visual, and LLM software in React, TypeScript, and Python. Shipped a live archive site for a DJ collective, an LLM-to-3D research pipeline, and a public LED data installation. Chicago, IL.

EDUCATION

DePaul University | B.S. UX Design, Minor in Computer Science | GPA: 3.835 Chicago, IL | June 2026

PROJECTS

MADMADHOUSE Archive and Booking CV | madmadhouse.com Sep 2026 – Oct 2026

- Turned a DJ collective's history, which lived only in Instagram flyers, into a live archive and booking CV (44 events, 21 venues, 3 cities) with scroll-driven motion (GSAP, Lenis) and posters that develop from ThumbHash blur-ups
- Built in Vite, React 19, and TypeScript on a single motion-token file; cut Cumulative Layout Shift at load from 0.113 to 0; validated the data pull by reconciling posts plus recorded failures against the source count; live on Cloudflare

  Vite, React 19, TypeScript, GSAP, Lenis, ThumbHash, Python, Cloudflare

LLM-to-Geometry Research Pipeline | DePaul Undergraduate Research Assistant Program Sep 2025 – Mar 2026

- Held the architecture constant and compared Mistral 7B with Claude Haiku, Sonnet, and Opus on the same descriptors; found the seven-parameter JSON schema, not the model alone, was the bottleneck, and redirected the work from prompt tuning to rendering architecture
- As sole developer, built a FastAPI server (Anthropic and Ollama provider registry, composite-key cache, SSE streaming) wired to TouchDesigner over WebSocket, then rebuilt and extended it as an independent public React Three Fiber demo in 2026 (urap-geometry.pages.dev)

  Python, FastAPI, Claude API, Ollama, TouchDesigner, React Three Fiber, SSE, procedural 3D geometry

Urban Displacement Data Visualization | Wabash Lights public installation, DePaul Spring 2025

- Designed a composite metric (renter-to-owner ratio shift) from Urban Displacement Project data across Chicago census tracts and mapped UDP's eight typologies to a light palette
- Built the p5.js animation that ran live on the Wabash Lights installation, working within the LED rail's constraints; sole designer, analyst, and developer

  p5.js, JavaScript, Tableau, composite metric design, color theory, data ethics, public installation

Caterpillar SSO and Access Control Redesign | DePaul Capstone, team of four Jan 2026 – Jun 2026

- Led the team's reframe of Caterpillar's login from a UI problem to a trust problem: field research at the demo and training center found operators defaulted to Guest Mode (adoption reported under 5%)
- Contributed to a three-layer architecture (identity, authorization, session) with NFC, SMS code, and PIN sign-in and a 10-flow hi-fi Figma prototype, reviewed in walkthroughs with Caterpillar UX and a Cat systems SME

  Figma, heuristic evaluation, field research, systems architecture, userflow mapping

SKILLS

Development: Python, JavaScript, TypeScript, HTML, CSS, React, React Three Fiber, Three.js, Vite, Node.js, FastAPI, SQL (Postgres), Git, p5.js, GSAP, WebSocket, SSE, TouchDesigner; Java, Scala

Design & Research: Figma, heuristic evaluation, semi-structured interviews, affinity mapping, personas, journey mapping, systems design, data visualization

AI / ML: LLMs (Anthropic Claude API, Ollama), prompt engineering, LLM evaluation, design tokens, procedural 3D geometry

CERTIFICATIONS

AI Fluency: Framework & Foundations (Anthropic) · Claude 101 (Anthropic) · Certified SolidWorks Associate, Mechanical Design (Dassault Systemes, 2020)